PY2105 — Assignment 4 Report

1 Linear systems

The system was solved using Gaussian elimination with partial pivoting and an unpivoted LU factorisation, with forward and backward substitution. No preconstructed matrix solver was used.

The upper-triangular matrix and transformed right-hand side after Gaussian elimination with partial pivoting were

$$U_{\mathrm{G}} = \begin{bmatrix} 13 & 4 & 7 & 9 \\ 0 & 13.076923 & 13.384615 & 8.923077 \\ 0 & 0 & -6.411765 & 10.058824 \\ 0 & 0 & 0 & -2.214679 \end{bmatrix}, \qquad b_{\mathrm{G}} = \begin{bmatrix} 111 \\ 137.384615 \\ 24.647059 \\ -11.073395 \end{bmatrix}.$$

Back substitution gives

$$x = \begin{bmatrix} 2 & 3 & 4 & 5 \end{bmatrix}^T, \qquad \det(A) = -2414.$$

The LU factorisation, satisfying $A = LU$, is

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0.769231 & 1 & 0 & 0 \\ 0.076923 & 2.631579 & 1 & 0 \\ 0.230769 & 4.473684 & 6.106383 & 1 \end{bmatrix},$$

$$U = \begin{bmatrix} 13 & 4 & 7 & 9 \\ 0 & 2.923077 & -0.384615 & 5.076923 \\ 0 & 0 & 2.473684 & 1.947368 \\ 0 & 0 & 0 & -25.680851 \end{bmatrix}.$$

Solving $Ly = b$ and then $Ux = y$ gives the same solution. The inverse, computed by solving $Az = e_j$ for each column e_j, is

$$A^{-1} \approx \begin{bmatrix} -0.197183 & 0.376139 & -0.163215 & -0.011599 \\ -0.781690 & 1.033140 & -0.384424 & 0.071665 \\ 0.521127 & -0.708368 & 0.217067 & 0.030655 \\ 0.338028 & -0.451533 & 0.237780 & -0.038940 \end{bmatrix}.$$

2 Monte Carlo estimate of π

Uniform points were sampled in a square of side $a = 2$, centred at the origin. The circle has radius $a/2 = 1$. If N_c of N_s points fall inside the circle, the estimate is

$$\widehat{\pi} = 4\frac{N_c}{N_s}.$$

For 10,000,000 points, the result was $\widehat{\pi} = 3.1419404$, with relative error 0.0110691%. The precision plot shows the measured relative error at increasing sample counts. Monte Carlo error is expected to scale statistically as $N^{-1/2}$; an individual run need not decrease monotonically, as seen in the plot.

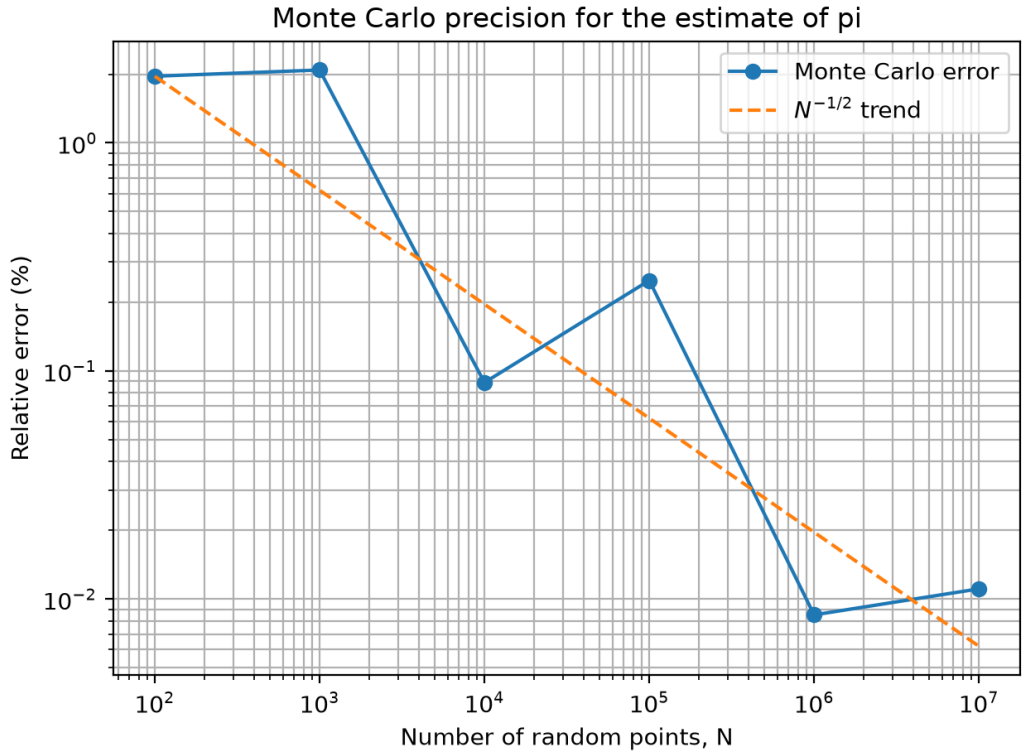


Figure 1: Relative error in the Monte Carlo estimate of π versus sample count. The dashed line shows an $N^{-1/2}$ reference trend.

3 Monte Carlo integration

The integrand is

$$f(x) = \frac{300x}{1 + e^{x}}, \qquad 0 \le x \le 10.$$

The bounding rectangle used for hit-or-miss sampling was $[0, 10] \times [0, 100]$. It encloses the function throughout the interval. The numerical reference integral, evaluated using the trapezoidal rule with one million intervals, was 246.5902935.

Using the binomial standard error to estimate a sample count for a 0.01% relative error at approximately 95% confidence gave 1,173,727,761 Monte Carlo points. One run returned 246.5882359, a relative error of 0.0008344% against the reference. The measured time was 248.04 s.

The trapezoidal rule required 226 intervals (step size $h = 10/226 \approx 0.0442478$) to obtain a relative error below 0.01% against the same reference. Its result was 246.5658002, with relative error 0.0099328%, and measured runtime 0.0000884 s. In this run, the Monte Carlo calculation took about 2.8×10^{6} times longer. Runtime depends on the machine and measurement conditions; the main computational difference is that the Monte Carlo estimate required over a billion samples.